Relational operators [== (comparison) , <, >, <=, >=, != (not equal)]:

They are used to check the given condition or expression is true or false. If condition true always the answer is 1 and false it is 0.

TC

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```c
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n",3==3);
printf("%d\n",9==7);
printf("%d\n",9==9.0);
printf("%d\n",0.90 == .9 );
printf("a addr %u\n","a");
printf("a addr %u\n","a");
printf("%d\n","a"=="a");
printf("%d\n",'a'==97);
printf("%d\n",'a'>'A');   /* 97 > 65 */
printf("%d\n",'a'<='A'+32);
printf("%d\n",7!=7);
printf("%d\n",7.01>7);
getch();
}
```


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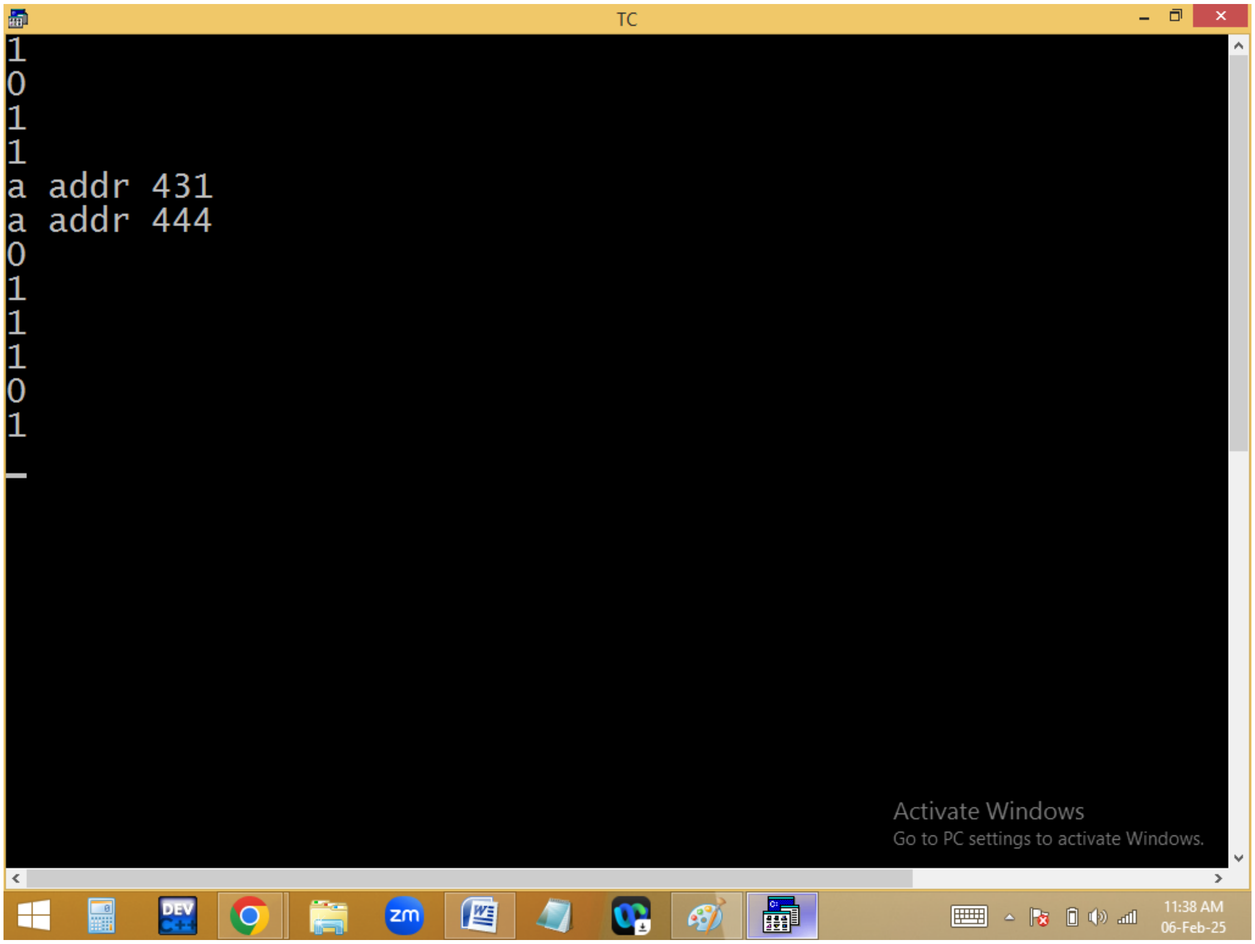
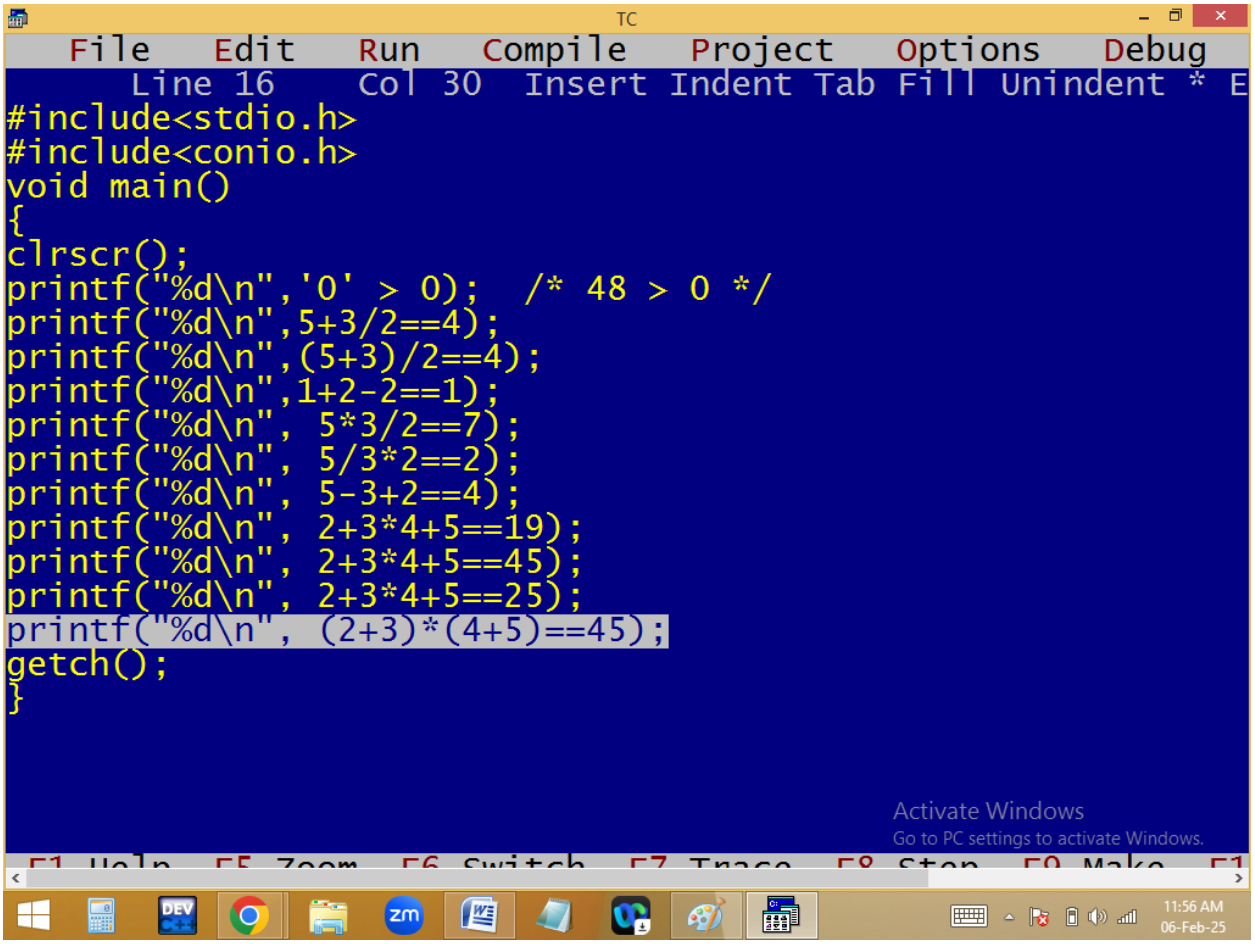


Operator precedence / Operator priority

(ASSOCIATION OF OPERATORS)

1. ( )
2. +, - , ! (sign operators, unary operators)
3. ++, -- (pre increment & decrement)
4. *, / , %
5. +, - (Binary)

6. = =, !=
7. &&
8. | |
9. **? :** (ternary operator)
10. =
11. ++, -- (Post increment & decrement)
12. , (comma)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Line 16 Col 30 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n",'0' > 0); /* 48 > 0 */
printf("%d\n",5+3/2==4);
printf("%d\n",(5+3)/2==4);
printf("%d\n",1+2-2==1);
printf("%d\n", 5*3/2==7);
printf("%d\n", 5/3*2==2);
printf("%d\n", 5-3+2==4);
printf("%d\n", 2+3*4+5==19);
printf("%d\n", 2+3*4+5==45);
printf("%d\n", 2+3*4+5==25);
printf("%d\n", (2+3)*(4+5)==45);
getch();
}
Activate Windows
Go to PC settings to activate Windows.
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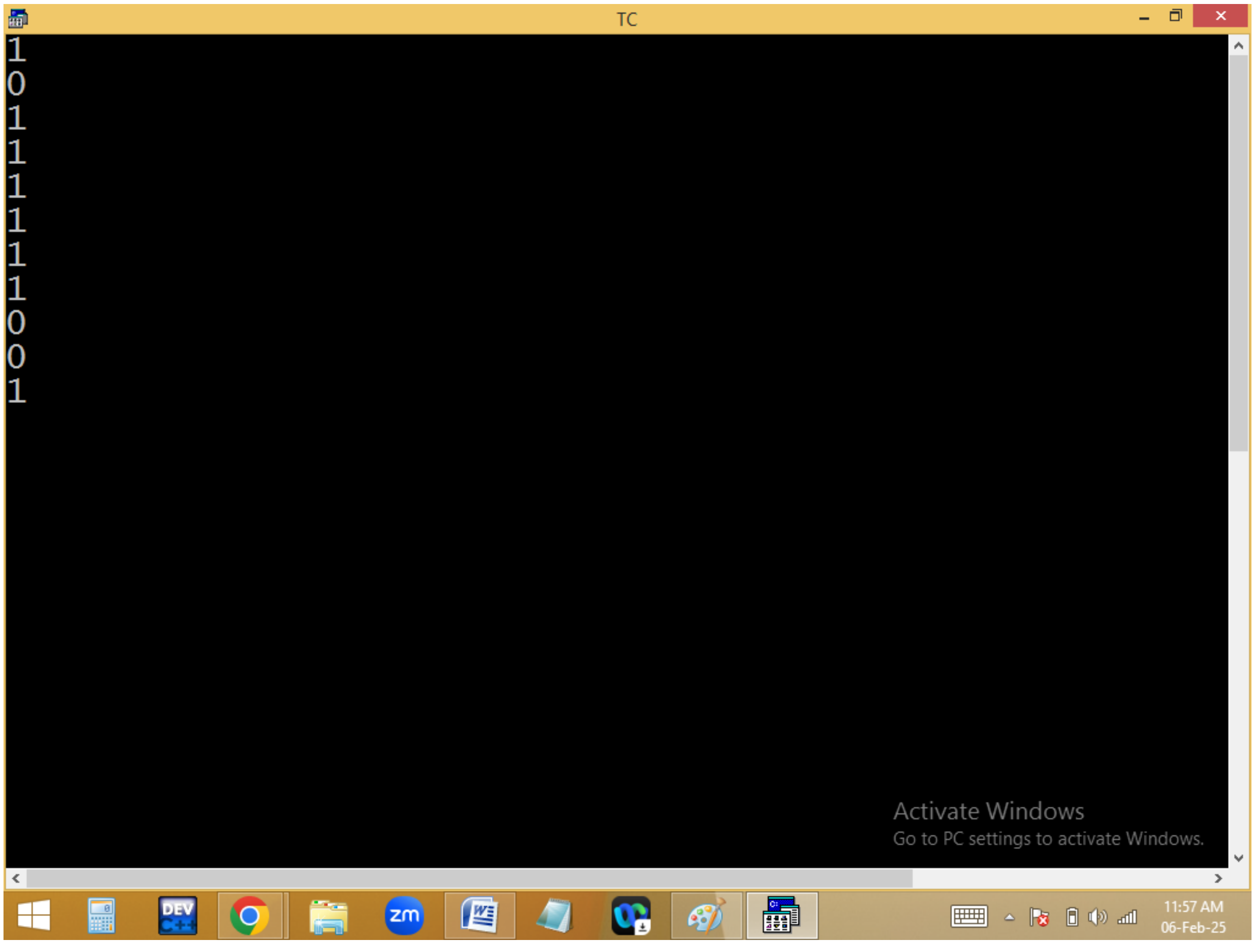
06-Feb-25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1
0
1
1
1
1
1
1
0
0
1
Activate Windows
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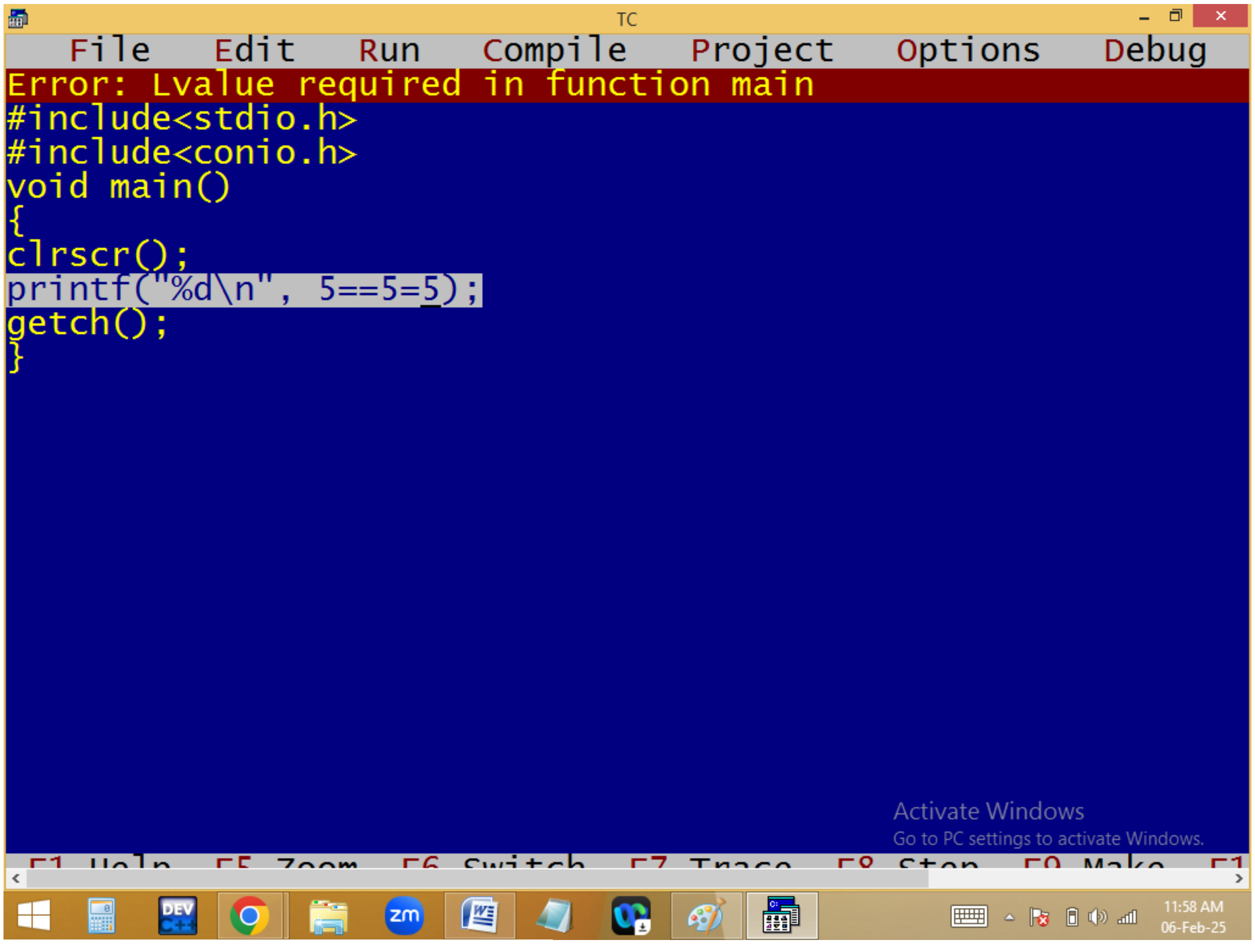
06-Feb-25



$$\underline{5 == 5} = 5$$

$$1 = 5$$

constant ==> Error

Logical operators:

1. && - Logical and
2. || - Logical or
3. ! – Logical not

&&, || used to combine two or more expressions into a single expression.

Not used for negation i.e. true becomes false and false become true.

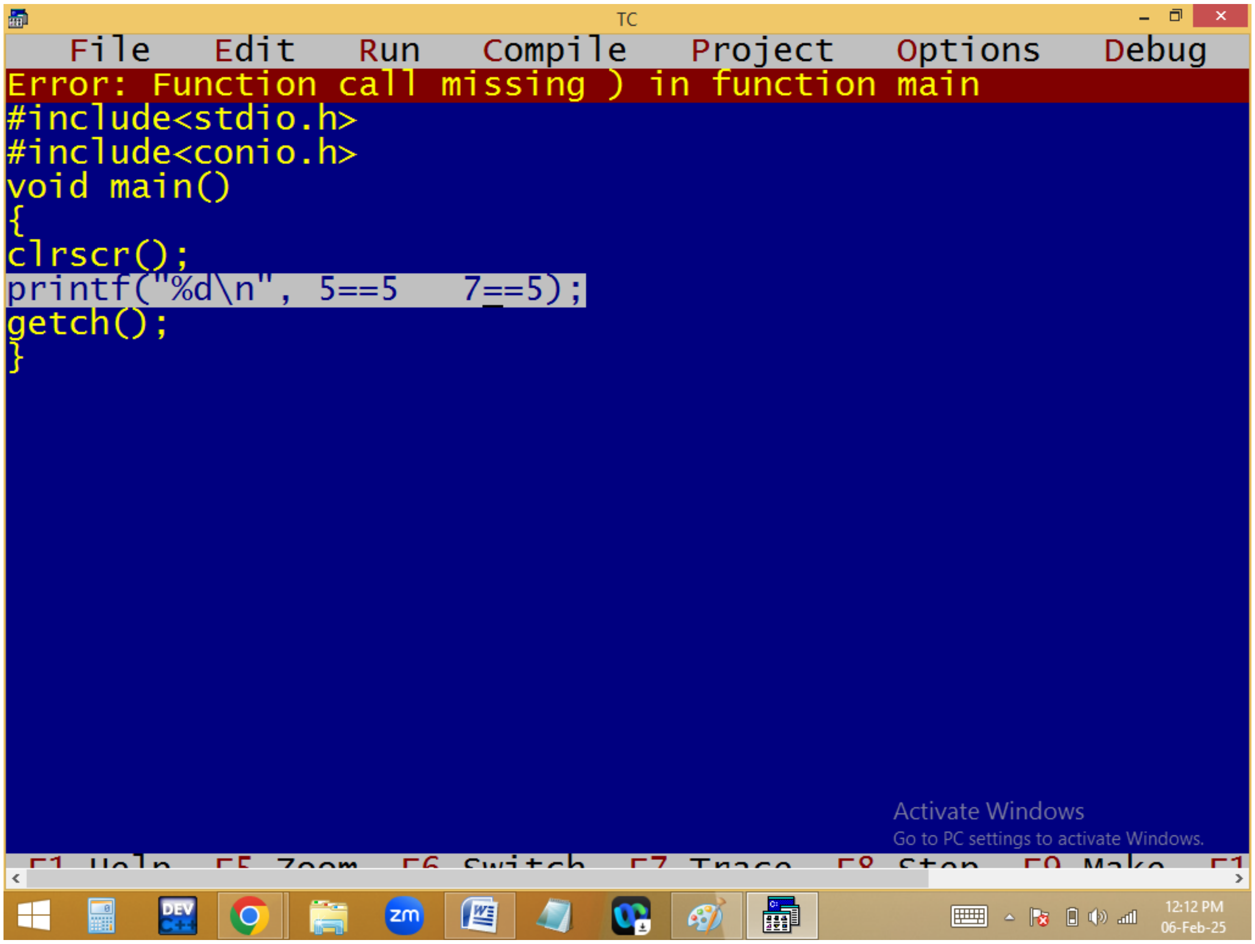
Note: In C && C++ 0 means false and other than 0 anything is true i.e. 1.

Truth tables:

Operator	Expression1	Expression2	Result
&&-and	1	1	1
	1	0	0
	0	1	0
	0	0	0
||-or	1	1	1
	1	0	1
	0	1	1
	0	0	0

!true = false

!false = true


TC
File Edit Run Compile Project Options Debug
Error: Function call missing) in function main
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n", 5==5 7==5);
getch();
}
Activate Windows
Go to PC settings to activate Windows.
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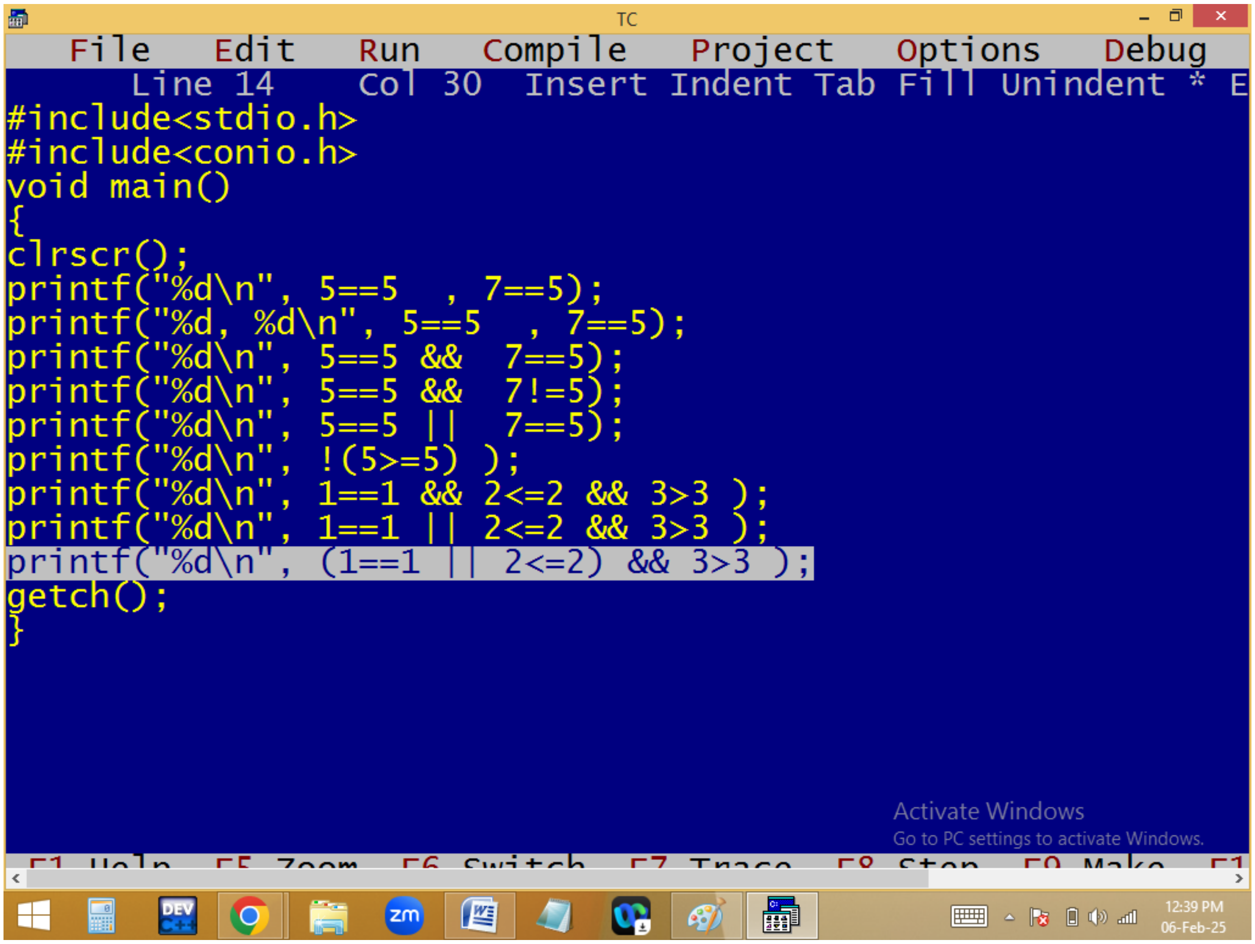

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#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n", 5==5 , 7==5);
printf("%d, %d\n", 5==5 , 7==5);
printf("%d\n", 5==5 && 7==5);
printf("%d\n", 5==5 && 7!=5);
printf("%d\n", 5==5 || 7==5);
printf("%d\n", !(5>=5));
printf("%d\n", 1==1 && 2<=2 && 3>3);
printf("%d\n", 1==1 || 2<=2 && 3>3);
printf("%d\n", (1==1 || 2<=2) && 3>3);
getch();
}
Activate Windows
Go to PC settings to activate Windows.
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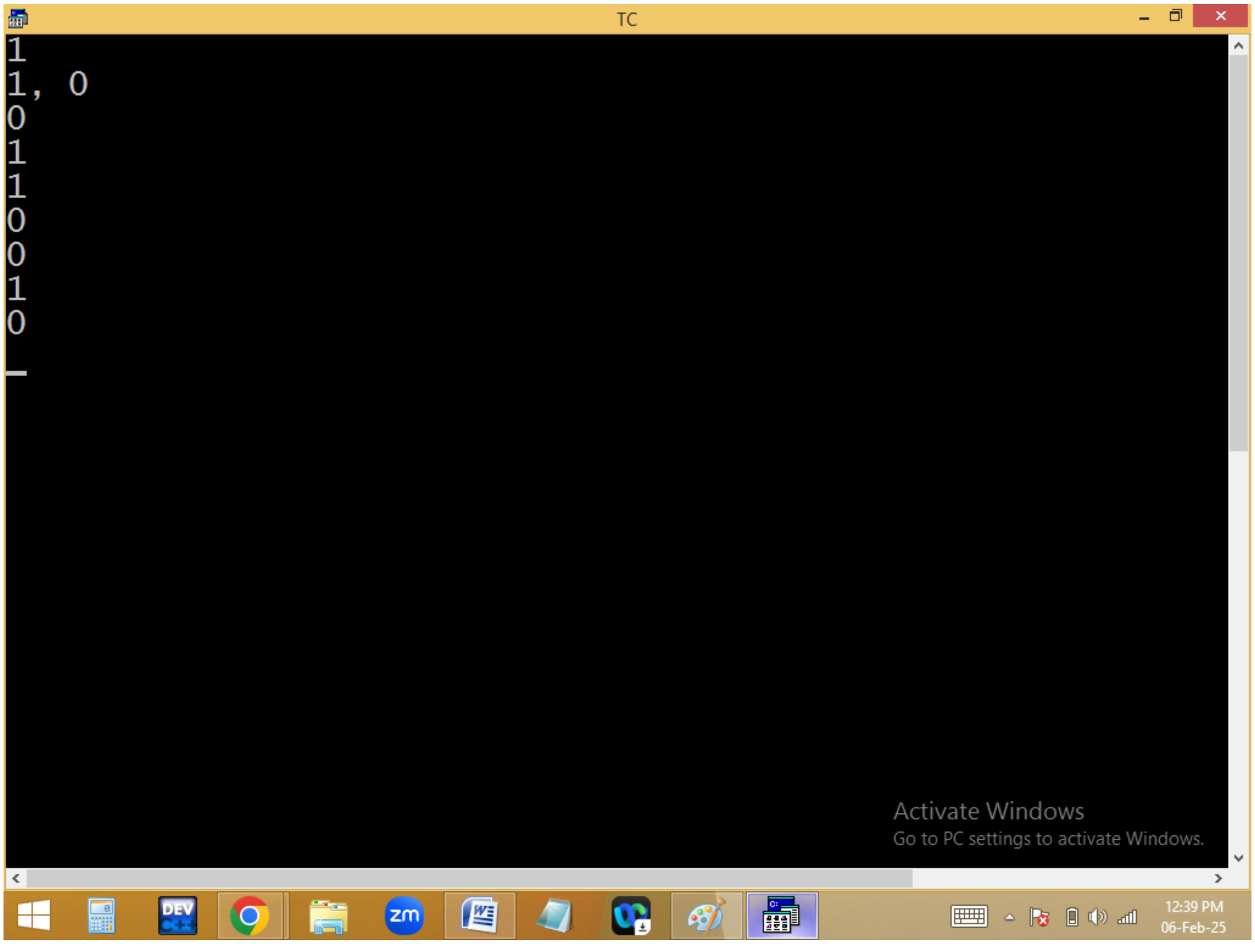


5==5 && 7!=5
1 && 1
1

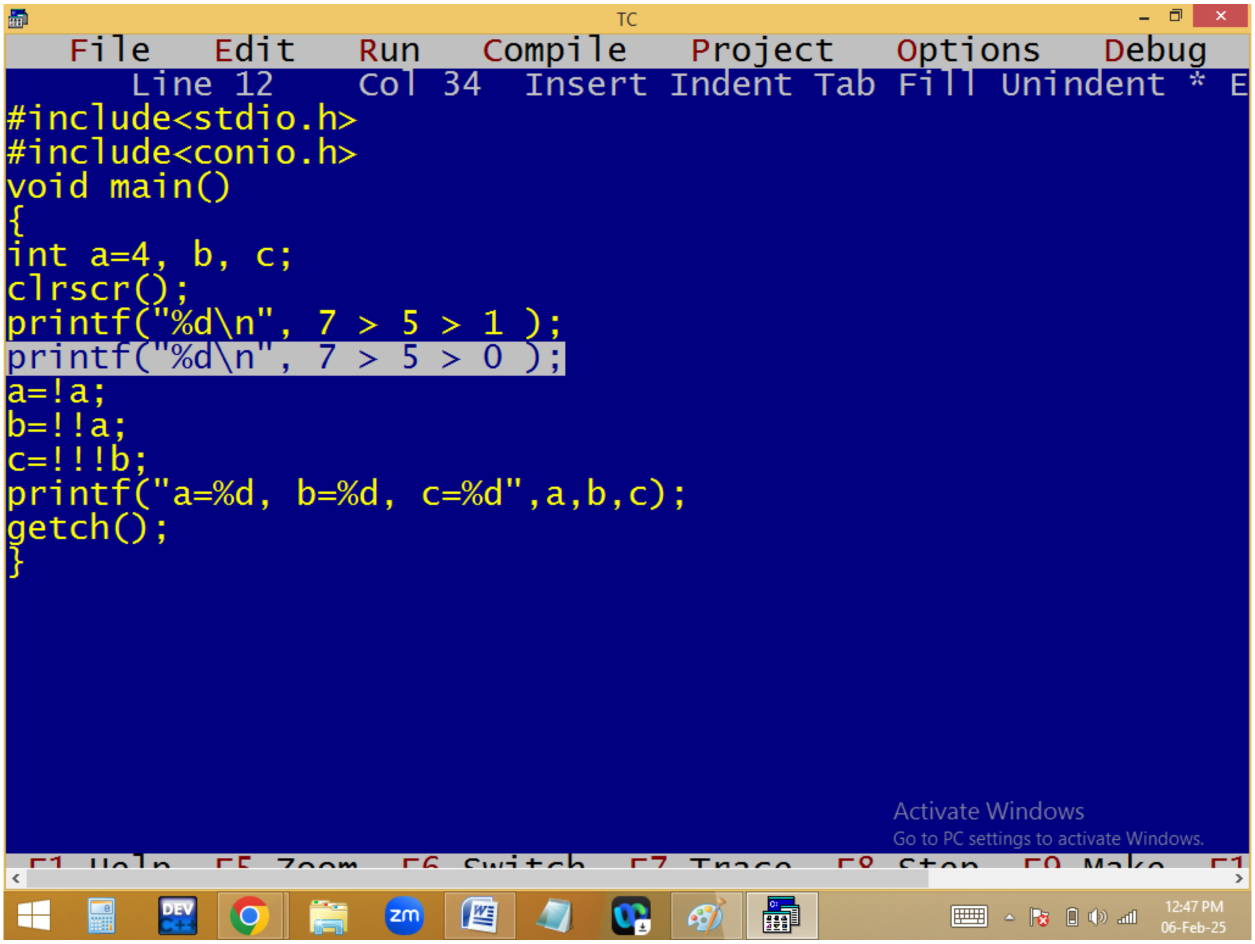
1!=1 || 6==9
0 || 0
0

!(7>7)
!0
1

1==1 && 2<=2 && 3>3
1 && 1
1 && 0
0

(1==1 || 2<=2) && 3>3
1 || 1
1 && 0
0

1==1 || 2<=2 && 3>3
1 && 0
1 || 0
1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Line 12 Col 34 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a=4, b, c;
clrscr();
printf("%d\n", 7 > 5 > 1);
printf("%d\n", 7 > 5 > 0);
a=!a;
b=!!a;
c=!!!b;
printf("a=%d, b=%d, c=%d",a,b,c);
getch();
}
Activate Windows
Go to PC settings to activate Windows.
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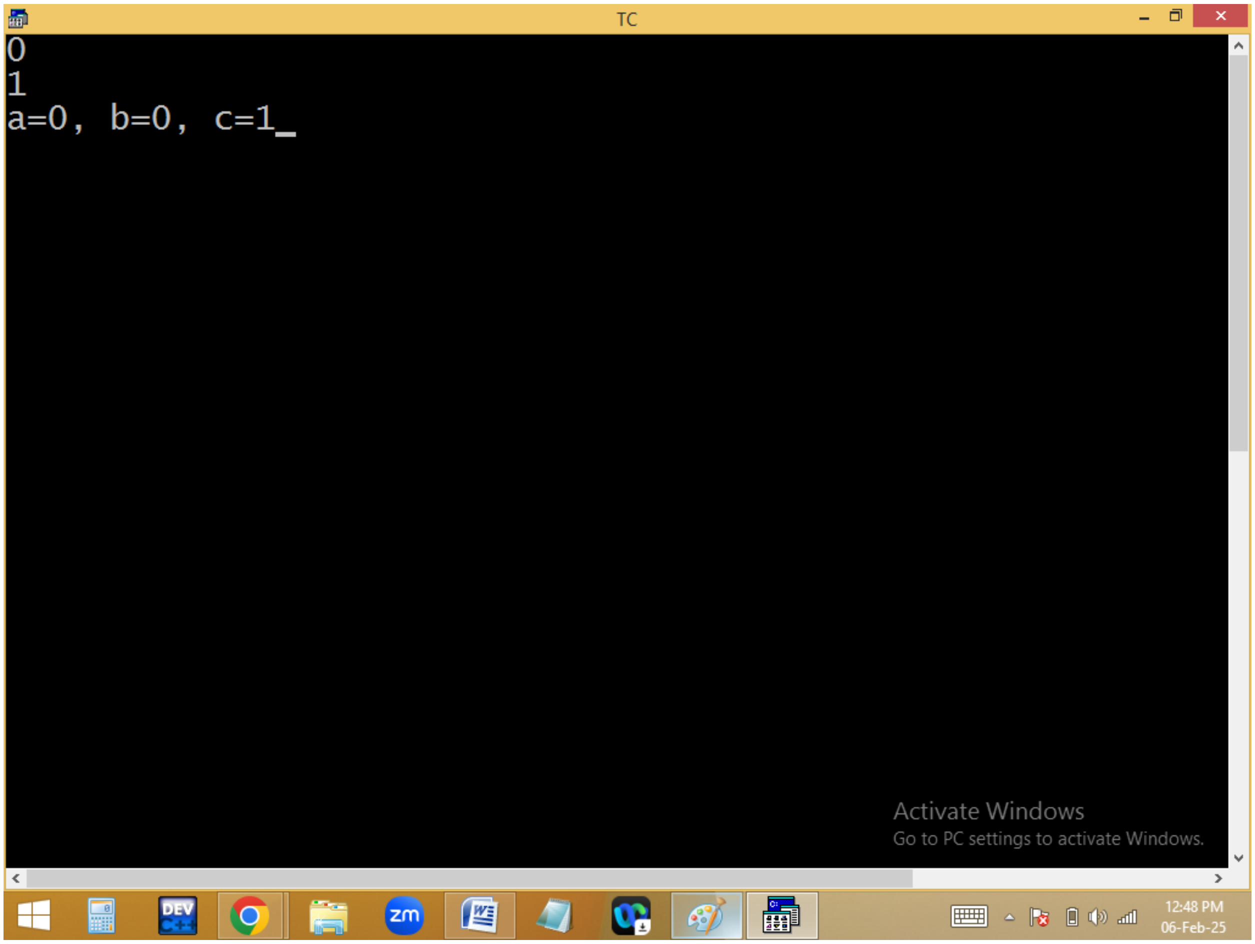

7 > 5 > 1

1 > 1

0

a=4

a = !a ==> !4 ==> !1 ==> a=0

b = !!a ==> !0 ==> 1 ==> !1 ==> b=0

c = !!!b ==> !0 ==> 1 ==> 0 ==> c=1